
PH 550 : Introduction to Quantum Optics

Course Code : PH 550

Course Name : Introduction to Quantum Optics

L-P-T-C: 3-0-0-3

Intended for: UG/PG/IPhD/PhD

Prerequisites: Basics of Quantum Mechanics , Classical Electrodynamics ,Mathematical Methods for Physics

Mutual Exclusion: None

Approval: 57th BoA

Course Contents

- **Module 1: Two-Level Atom System** (8 lectures)
 - Oscillating dipoles and Bloch sphere representation
 - Density Operator and Density Matrix formalism
 - Optical Bloch Equations
 - Rabi Frequency and Spontaneous Decay
 - Lineshape in Fluorescence and Saturation Broadening
- **Module 2: Quantization of the Electromagnetic Field** (14 lectures)
 - Quantized Modes and Operators
 - Ladder Operators, Number Operator, and Quadrature Operators
 - Quantization of the Electromagnetic Field
 - Hamiltonian of Radiation Field and Vacuum State Energy
 - Fock States, Coherent States, and Squeezed States
 - Thermal States and Planck's Black Body Radiation Formula
- **Module 3: Photon Statistics and Measurement** (10 lectures)
 - Photon Statistics
 - * Poissonian, Sub-Poissonian, and Super-Poissonian light
 - * Intensity interferometer and Hanbury Brown–Twiss experiments
 - * Second-order correlation function, $g_2(t)$
 - * Photon bunching and antibunching phenomena
 - * Coherent, Bunched, and Antibunched light
 - Experimental Demonstrations
 - * Photon antibunching experiments
 - * Single-photon sources and their application in quantum technologies
- **Module 4: Atoms in Cavities** (10 lectures)
 - Optical and Microwave Cavities
 - * Fundamentals of Optical Cavities

- * Microwave Cavities and Rydberg Atoms
- Atom-Cavity Coupling
 - * Weak coupling regime and the Purcell Effect
 - * Experimental observations of the Purcell Effect
 - * Strong coupling regime and Cavity QED
 - * Experimental observation of strong coupling in cavities

Laboratory:

1. NA

Text books:

1. Scully, M. O., & Zubairy, M. S., **Quantum optics**.
2. Christopher Gerry and Peter Knight, **Introductory Quantum Optics**.
3. Mark Fox, **Quantum Optics**.
4. Leonard Mandel and Emil Wolf, **Optical Coherence and Quantum Optics**.

References:

1. NA